

Part A – Theoretical Foundation

1. What is inferential statistics?

→ Inferential Statistics:

Inferential statistics means using data from a sample to make conclusions about the whole population. It helps us predict or estimate values without checking everyone.

◆ Example:

Studying 200 patients to estimate the average blood pressure of all people in a city.

2. What is hypothesis testing and its components?

→ Hypothesis Testing:

Hypothesis testing is a method to check if a claim about a population is true.

Null hypothesis (H_0): No effect or no difference.

Alternative hypothesis (H_1): There is an effect or difference.

◆ Example:

H_0 : “Smoking does not affect diabetes.”

H_1 : “Smoking affects diabetes.”

3. Explain confidence interval and critical value.

→ Confidence Interval :

A confidence interval gives a range where the true value is likely to lie, with a certain confidence (like 95%).

◆ Example:

The average BMI is 24, and the 95% confidence interval is [23,

25], meaning the true average is likely between 23 and 25.

→ Critical Value:

The critical value is the cut-off point used to decide whether to accept or reject H_0 .

◆ **Example:**

Average BMI = 24, CI = [23, 25]. If test statistic > critical value, we reject H_0 .

4. Define p-value.

→ p-value:

The p-value shows the probability that the observed result happened by chance if H_0 is true.

◆ **Example:**

$p = 0.03$ means only 3% chance of the result is random. If $p < 0.05$, we reject H_0 .

5. Differentiate Type I and Type II errors.

→ Type I Error: Rejecting H_0 when it is true (false alarm).

◆ **Example:** Saying a healthy person has diabetes.

→ Type II Error: Not rejecting H_0 when it is false (missed detection).

◆ **Example:** Saying a diabetic person is healthy.

6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

→ **Z-test:**

Used for large samples ($n > 30$) or when population variance is known.

- ◆ **Example:** Compare average blood pressure of 500 men vs 500 women.

→ **T-test:**

Used for small samples ($n < 30$) or when variance is unknown.

- ◆ **Example:** Compare the average BMI of 20 smokers vs 20 non-smokers.

→ **Chi-square test:**

Compares observed vs expected frequencies in categorical data.

- ◆ **Example:** Check if smoking habit is linked to diabetes.

→ **ANOVA:**

Compares means across more than two groups by splitting variance into between-group and within-group parts.

- ◆ **Example:** Compare cholesterol levels across 5 age groups.

7. What is Covariance?

→ **Covariance:** Covariance shows how two variables change together.

- ◆ **Example:** If age increases and BMI also increases, covariance is positive. If one increases while the other decreases, covariance is negative.
- ◆ **Limitation:** Value depends on units, so correlation is often used instead.

8. What is Correlation?

→ Correlation:

Correlation is a scaled version of covariance that shows both strength and direction of the relationship between two variables.

- ◆ **Example:** Correlation = 0.9 between age and BMI means a strong positive relationship (as age increases, BMI also increases).